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Department of Computer Science and Engineering

Unit – 4 Network Layer and Internet Protocol

4.1 Overview of Network Layer

4.2 What's Inside a Router?

4.3 Switching

4.4 The Internet Protocol (IP)

- Datagram format
- Fragmentation
- IPv4 addressing
- NAT

4.5 Introduction to Network Layer Protocols

- DHCP
- ICMP

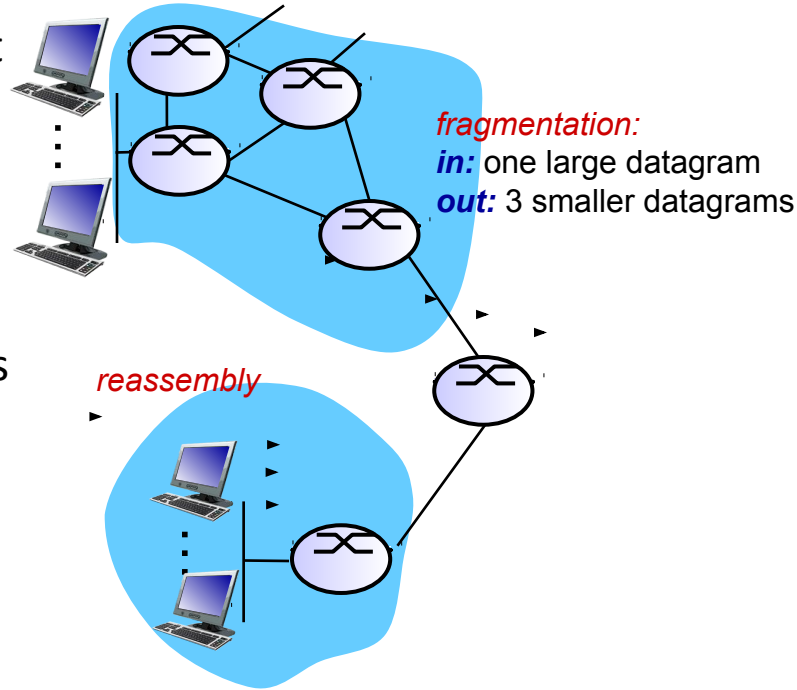
4.6 IPv6 Addressing

4.7 Introduction to Routing Algorithms

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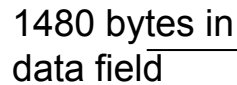
IP fragmentation, reassembly

- network links have MTU (max.transfer size) - largest possible link-level frame
 - different link types, different MTUs
- large IP datagram divided (“fragmented”) within net
 - one datagram becomes several datagrams
 - “reassembled” only at final destination
 - IP header bits used to identify, order related fragments



- ❖ 4000 byte datagram
- ❖ MTU = 1500 bytes

one large datagram becomes
several smaller datagrams



offset

$$=$$

1480/8

length	ID	fragfla	offset
=4000	=x	g	=0
		=0	

	length	ID	fragfla	offset
	=1500	=x	g	=0
			=1	

	length	ID	fragfla	offset
	=1500	=x	g	=185
			=1	

	length	ID	fragfla	offset
	=1040	=x	g	=370
			=0	

Original IP Datagram

Sequence	Identifier	Total Length	DF May / Don't	MF Last / More	Fragment Offset
0	345	5140	0	0	0

IP Fragments (Ethernet)

Sequence	Identifier	Total Length	DF May / Don't	MF Last / More	Fragment Offset
0-0	345	1500	0	1	0
0-1	345	1500	0	1	185
0-2	345	1500	0	1	370
0-3	345	700	0	0	555

Fragmentation – Numerical Example

- An IP router with a Maximum Transmission Unit (MTU) of 200 bytes has received an IP packet of size 520 bytes with an IP header of length 20 bytes. The values of the relevant fields in the IP header.





THANK YOU

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